

Bioinfo Assignment 3

Bohan YANG
Student ID: 2330016056

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Question 1

Let $P = (p_{xy})_{x,y \in \{A,C,G,T\}}$ be the given 4×4 transition matrix (rows and columns in the order A, C, G, T). The root sequence is fixed as

$$X_0 = AAA,$$

so at each site $k = 1, 2, 3$ the root state is

$$R_k = A.$$

The observed sequences are

$$X_1 = AGT, \quad X_2 = AAC, \quad X_3 = AGG.$$

Because sites evolve i.i.d., the likelihood of a tree topology T is

$$L(T) = \prod_{k=1}^3 L_k(T), \quad \ell(T) = \log L(T) = \sum_{k=1}^3 \log L_k(T),$$

where $L_k(T)$ is the likelihood for site k .

Let $H_k \in \{A, C, G, T\}$ be the (unobserved) state at the internal node h at site k . Using the tree structures, the per-site likelihoods are:

Tree 1: $0 \rightarrow h, h \rightarrow 1, h \rightarrow 2, 0 \rightarrow 3$

$$L_k(T_1) = p_{A,X_{3k}} \sum_h p_{A,h} p_{h,X_{1k}} p_{h,X_{2k}}.$$

Tree 2: $0 \rightarrow h, h \rightarrow 3, h \rightarrow 2, 0 \rightarrow 1$

$$L_k(T_2) = p_{A,X_{1k}} \sum_h p_{A,h} p_{h,X_{3k}} p_{h,X_{2k}}.$$